

1 Foundations for the convergence proof

We first establish two facts used throughout the proof. A policy that chooses actions at least as good as those of a reference policy improves its values. Also, sampled updates remain bounded and converge whenever their policy-evaluation target eventually stops changing. The second fact needs infinite accumulated learning at each pair, rather than a positive lower bound on its sampling probability.

1.1 Values and policy improvement

Let $\mathcal{S}$ be a finite set of nonterminal states, with finite nonempty action sets $\mathcal{A}(s)$, and write

$$\mathcal{I} = \{(s, a) : s \in \mathcal{S}, a \in \mathcal{A}(s)\}, \quad J = |\mathcal{I}|.$$

Terminal states have value zero and are omitted from $\mathcal{I}$. One-step rewards have finite expected absolute value. The set $\mathcal{P}$ consists of deterministic stationary policies. We assume either $0 \leq \gamma < 1$, or $\gamma = 1$ and every policy in $\mathcal{P}$ reaches a terminal state almost surely from every state. Such a policy is called proper.

The state value $v_\pi(s)$ is the expected discounted return starting at s and following π . The action value $q_\pi(s, a)$ prescribes the initial action a and follows π thereafter. Let $p(s' | s, a)$ be the transition probability and $r(s, a)$ the expected immediate reward. Then

$$q_\pi(s, a) = r(s, a) + \gamma \sum_{s' \in \mathcal{S}} p(s' | s, a) v_\pi(s'), \quad v_\pi(s) = q_\pi(s, \pi(s)). \quad (1)$$

All vector inequalities are coordinatewise and $\|\cdot\|_\infty$ denotes the maximum absolute coordinate. Policy order $\pi' \succeq \pi$ means $v_{\pi'} \geq v_\pi$, which implies $q_{\pi'} \geq q_\pi$ by (1). The converse need not hold: taking a prescribed initial action can bypass the only decision at which the policies differ.

We record why the episodic assumption suffices even when comparing a stationary policy with arbitrary control laws.

Lemma 1 (Uniform termination) *If every deterministic stationary policy is proper, there exist an integer $m \geq 1$ and $\rho < 1$ such that the termination time T satisfies $\mathbb{P}(T > jm) \leq \rho^j$ for every $j \geq 0$, uniformly over initial states, prescribed initial actions and history-dependent control laws. In particular, $\sup \mathbb{E}[T^2] < \infty$.*

Proof. Let $w_n(s)$ be the largest probability of remaining nonterminal after n transitions. Conditioning on the first transition gives

$$w_0(s) = 1, \quad w_{n+1}(s) = \max_{a \in \mathcal{A}(s)} \sum_{s'} p(s' | s, a) w_n(s').$$

These probabilities decrease to a vector w . Since the sets are finite, the limit satisfies the same equation. Choose a stationary maximizing policy π in this equation, and denote its nonterminal transition matrix by P_π . Then $w = P_\pi w = P_\pi^n w$. Properness implies $P_\pi^n \rightarrow 0$, so $w = 0$. Choose m with $\max_s w_m(s) \leq \rho < 1$. Conditioning after each block of m transitions proves the bound. Prescribing the first action cannot increase the maximum. Summing this geometric tail gives the second-moment bound. $\blacksquare$

Values are finite under either discounting or properness. Since $\mathcal{P}$ is finite, so is

$$B = \max_{\pi \in \mathcal{P}} \|q_\pi\|_\infty. \quad (2)$$

Lemma 2 (Policy improvement) *If $\pi, \pi' \in \mathcal{P}$ satisfy*

$$q_\pi(s, \pi'(s)) \geq q_\pi(s, \pi(s)) \quad (s \in \mathcal{S}), \quad (3)$$

then $v_{\pi'} \geq v_\pi$ and $q_{\pi'} \geq q_\pi$. If $q_{\pi'} \neq q_\pi$, both vector comparisons are strict in at least one coordinate. A policy greedy for its own action values is optimal, including among history-dependent control laws. An optimal deterministic stationary policy π_ exists.*

Proof. Write $r_{\pi'}(s) = r(s, \pi'(s))$ and $P_{\pi'}(s, s') = p(s' | s, \pi'(s))$. Set

$$h = r_{\pi'} + \gamma P_{\pi'} v_\pi - v_\pi \geq 0.$$

Subtracting the value equations gives

$$v_{\pi'} - v_\pi = \sum_{j=0}^{\infty} (\gamma P_{\pi'})^j h \geq 0. \quad (4)$$

The series converges under discounting or properness. Equation (1) gives the action-value comparison. If the state values were equal, that equation would make the action values equal as well, proving the strictness assertion.

If π is greedy for q_π , then for every action a ,

$$v_\pi(s) \geq r(s, a) + \gamma \sum_{s'} p(s' | s, a) v_\pi(s').$$

Iterating this inequality along any control law bounds its expected return through time n by v_π plus a remainder of absolute value at most $\|v_\pi\|_\infty \gamma^n$ in the discounted case, or $\|v_\pi\|_\infty \mathbb{P}(T > n)$ in the undiscounted case. The remainder tends to zero, using Lemma 1 in the latter case. Finite expected absolute rewards and the same tail bounds justify passing to the full return. Policy π attains v_π , so it is optimal.

Finally choose π maximizing $\sum_s v_\pi(s)$ over the finite set $\mathcal{P}$. If it were not greedy for q_π , a greedy policy π' would make $h \geq 0$ with a positive coordinate. The $j = 0$ term in (4) would strictly increase the sum, a contradiction. All optimal policies share the values denoted by v_{π_*} and q_{π_*} . $\blacksquare$

1.2 Initial-visit updates and information

Let $\mathcal{Q} = \mathbb{R}^{\mathcal{I}}$. For $q \in \mathcal{Q}$, retain the manuscript's notation

$$P(q) = \{\pi \in \mathcal{P} : q(s, \pi(s)) = \max_a q(s, a) \text{ for every } s\}, \quad Q(\pi) = \{q \in \mathcal{Q} : \pi \in P(q)\}.$$

MC-O-PI starts from a finite deterministic table q_0 . At iteration k it selects $\pi_k \in P(q_k)$, draws the initial pair I_{k+1} according to σ_k , obtains a return G_{k+1} by following π_k after that pair, and updates only this pair:

$$q_{k+1}(i) = q_k(i) + \alpha_k u_k(i) (q_{\pi_k}(i) + v_k(i) - q_k(i)), \quad i \in \mathcal{I}, \quad (5)$$

where $u_k(i) = \mathbf{1}_{\{I_{k+1}=i\}}$. On the sampled pair, $v_k(i) = G_{k+1} - q_{\pi_k}(i)$; set $v_k(i) = 0$ otherwise. For ordinary episodic returns the policy remains fixed until termination.

The history $\mathcal{F}_k$ contains the information available after q_k is computed but before selecting π_k and σ_k . Write $\mathcal{F}'_k = \sigma(\mathcal{F}_k, \pi_k, \sigma_k)$ for the post-selection history. Thus

$$\mathcal{F}_k \subseteq \mathcal{F}'_k \subseteq \sigma(\mathcal{F}'_k, I_{k+1}) \subseteq \mathcal{F}_{k+1}.$$

A history is represented mathematically by a sigma-algebra, and conditional expectation averages over the remaining randomness given that information. An increasing sequence of histories is a filtration. All stopping times below refer to $(\mathcal{F}'_k)$: whether such a time has occurred by k can be decided from $\mathcal{F}'_k$.

Assumption 3 (Standing assumptions) *The deterministic steps satisfy $0 < \alpha_k \leq 1$ and $\sum_k \alpha_k^2 < \infty$. The sampling law may depend on the entire post-selection history, but every pair receives infinite conditional learning weight:*

$$\sum_{k=0}^{\infty} \alpha_k \sigma_k(i) = \infty \quad \text{almost surely for every } i \in \mathcal{I}. \quad (6)$$

For a deterministic constant $c_v < \infty$, on every positive-probability pair event,

$$\mathbb{E}[G_{k+1} \mid \mathcal{F}'_k, I_{k+1} = i] = q_{\pi_k}(i), \quad \mathbb{E}[(G_{k+1} - q_{\pi_k}(i))^2 \mid \mathcal{F}'_k, I_{k+1} = i] \leq c_v. \quad (7)$$

No conditional law is required on a zero-probability pair event.

Condition (6) implies $\sum_k \alpha_k = \infty$, giving the usual Robbins–Monro conditions. Infinite visitation alone would not suffice: infinitely many updates can still have finite total learning weight. Bounded one-step rewards ensure (7) for ordinary episode returns, by discounting or Lemma 1; otherwise we explicitly assume these return moments.

For initial visits, the update distribution is $\mu_k(i) = \mathbb{E}[u_k(i) \mid \mathcal{F}'_k] = \sigma_k(i)$. The combined noise and mean-field representation are

$$w_k(i) = u_k(i)v_k(i) + (u_k(i) - \mu_k(i))(q_{\pi_k}(i) - q_k(i)), \quad (8)$$

$$q_{k+1} = q_k + \alpha_k(\mu_k(q_{\pi_k} - q_k) + w_k), \quad (9)$$

with coordinatewise products. The moments in (7) imply

$$\mathbb{E}[w_k(i) \mid \mathcal{F}'_k] = 0, \quad \mathbb{E}[w_k(i)^2 \mid \mathcal{F}'_k] \leq \sigma_k(i)\{c_v + (B + \|q_k\|_{\infty})^2\}. \quad (10)$$

Indeed, $w_k(i)$ is the centered version of $u_k(i)(G_{k+1} - q_k(i))$, and variance is bounded by its second moment. Fixed-time tables have finite second moments, by induction in (5).

1.3 Accumulated noise and persistent targets

A martingale difference is a random increment whose conditional mean given the previous history is zero. We use the following elementary consequence of square summability: if such increments e_{k+1} satisfy $\sum_k \mathbb{E}[e_{k+1}^2] < \infty$, their partial sums converge almost surely. To see

the tail control behind this fact, orthogonality of distinct increments and the martingale maximal inequality give

$$\mathbb{P} \left(\sup_{n \geq r} \left| \sum_{k=r}^{n-1} e_{k+1} \right| > \varepsilon \right) \leq \varepsilon^{-2} \sum_{k=r}^{\infty} \mathbb{E}[e_{k+1}^2].$$

For a finite horizon this inequality follows by stopping at the first threshold crossing: the final squared sum has expectation at least the squared stopped sum on the crossing event. Letting the horizon increase proves the display. Choose r_j making the variance tail at most 2^{-3j} and take $\varepsilon = 2^{-j}$; the probabilities then sum to a finite number. The elementary Borel–Cantelli implication (events whose probabilities sum to a finite number occur only finitely often) shows that these tails are uniformly small eventually. Hence the partial sums are Cauchy almost surely.

Lemma 4 (Actual learning weight) *Under the standing assumptions, almost surely*

$$\sum_k \alpha_k u_k(i) = \infty \quad (i \in \mathcal{I}).$$

Proof. The increments $\alpha_k(u_k(i) - \sigma_k(i))$ have conditional mean zero given $\mathcal{F}'_k$ and second moment at most α_k^2 . Their series converges by the preceding fact. Thus the partial sums of actual and conditional learning weights differ by a convergent sequence. Equation (6) proves the claim. Intersect the probability-one events over the finitely many pairs. ■

Lemma 5 (Filtering accumulated errors) *Consider a scalar recursion $x_{k+1} = (1 - b_k)x_k + b_k y_k + e_{k+1}$ with $0 \leq b_k \leq 1$. For $n > r$,*

$$\left| \sum_{k=r}^{n-1} e_{k+1} \prod_{j=k+1}^{n-1} (1 - b_j) \right| \leq 2 \max_{r \leq m \leq n} \left| \sum_{k=r}^{m-1} e_{k+1} \right|. \quad (11)$$

If $\sum_k b_k = \infty$ and $\sum_k e_{k+1}$ converges, eventual bounds $\ell \leq y_k \leq h$ imply $\ell \leq \liminf_k x_k \leq \limsup_k x_k \leq h$. In particular, an eventually constant target y gives $x_k \rightarrow y$. These are pathwise statements, with no independence assumption.

Proof. Unrolling the recursion assigns the initial value weight $\prod_{k=r}^{n-1} (1 - b_k)$, the targets nonnegative weights summing to one minus this product, and the errors the weights in (11). Put $E_m = \sum_{k=r}^{m-1} e_{k+1}$ and $a_k = \prod_{j=k+1}^{n-1} (1 - b_j)$. Summation by parts gives

$$\sum_{k=r}^{n-1} (E_{k+1} - E_k) a_k = E_n a_{n-1} - \sum_{k=r+1}^{n-1} E_k (a_k - a_{k-1}).$$

Here $0 \leq a_r \leq \dots \leq a_{n-1} = 1$, proving (11). Since $\sum_k b_k = \infty$, the initial weight tends to zero for fixed r . Convergence of the error series makes its tail partial sums uniformly small. Take r after the target bounds begin, let $n \rightarrow \infty$, and then let $r \rightarrow \infty$. ■

Lemma 6 (Stability and persistent targets) *Under the standing assumptions, on one probability-one event,*

$$\limsup_k \|q_k\|_\infty \leq B.$$

On the same event, if $q_{\pi_k} = \bar{q}$ for all sufficiently large k , then $q_k \rightarrow \bar{q} = q_{\pi_}$.*

Proof. For a pair i , use $b_k = \alpha_k u_k(i)$, $y_k = q_{\pi_k}(i)$ and $e_{k+1} = \alpha_k u_k(i) v_k(i)$ in Lemma 5. The weights sum to infinity by Lemma 4. The error increments have conditional mean zero and second moment at most $c_v \alpha_k^2$, so their series converges almost surely. Intersect these events over all pairs. The targets lie in $[-B, B]$, giving the asserted bound, and an eventual target $\bar{q}$ gives $q_k \rightarrow \bar{q}$.

Choose a policy $\hat{\pi}$ occurring infinitely often after the target has become constant. It satisfies $q_{\hat{\pi}} = \bar{q}$. Along that subsequence its greedy inequalities pass to the limit, so it is greedy for its own action values. Policy improvement and verification therefore give $\bar{q} = q_{\pi_*}$. This argument is pathwise; it does not condition on the event that the target eventually becomes constant. $\blacksquare$

A transition time records a change of the target, not merely a change of policy label. Starting at $t_0 = 0$, define

$$t_{n+1} = \inf\{k > t_n : q_{\pi_k} \neq q_{\pi_{t_n}}\} \quad (t_n < \infty), \quad (12)$$

with $\inf \emptyset = \infty$ and $t_{n+1} = \infty$ when $t_n = \infty$. These are stopping times for $(\mathcal{F}'_k)$. On $[t_n, t_{n+1})$ the target is constant even if the selected policy changes. To prove convergence it therefore suffices to prove that only finitely many transitions occur. A nonoptimal policy can share the optimal action-value table, so its selection alone does not imply a future target change.

2 Mean-field policy improvement

Averaging the sampled update over its pair and return gives

$$q_{k+1}(s, a) = q_k(s, a) + \alpha_k \mu_k(s, a) (q_{\pi_k}(s, a) - q_k(s, a)), \quad \pi_k \in P(q_k). \quad (13)$$

For this deterministic recursion, impose $0 < \alpha_k \leq 1$, $0 \leq \mu_k(s, a) \leq 1$ and uniform updates within each state:

$$\mu_k(s, a) = \mu_k(s) \quad (a \in \mathcal{A}(s)), \quad \sum_k \alpha_k \mu_k(s) = \infty \quad (s \in \mathcal{S}). \quad (14)$$

The scalar $\mu_k(s)$ denotes the probability of each pair at s , not the state marginal. Use full inertia: choose π_0 greedily, and thereafter

$$\pi_{k+1}(s) = \pi_k(s) \quad \text{whenever } \pi_k(s) \in \arg \max_a q_{k+1}(s, a). \quad (15)$$

If the preceding action is no longer greedy, any maximizing action may be selected.

Proposition 7 *Under these conditions $v_{\pi_{k+1}} \geq v_{\pi_k}$ and $q_{\pi_{k+1}} \geq q_{\pi_k}$ for every k . At every finite target transition the state values improve strictly in at least one state.*

Proof. Fix s and put $c = \alpha_k \mu_k(s)$. If $c > 0$, greediness before and after the update gives

$$q_{\pi_k}(s, \pi_{k+1}(s)) - q_{\pi_k}(s, \pi_k(s)) \geq \frac{1-c}{c} (q_k(s, \pi_k(s)) - q_k(s, \pi_{k+1}(s))) \geq 0.$$

If $c = 0$, the table at s is unchanged and full inertia retains the action, giving equality. Lemma 2 proves both comparisons. Across a target transition they remain valid by transitivity, and distinct action-value targets force a strict state-value comparison. ■

Proposition 8 *Under the same conditions, $q_k \rightarrow q_{\pi^*}$. This conclusion needs neither square summability nor step-size regularity.*

Proof. A nondecreasing sequence in the finite set $\{q_\pi : \pi \in \mathcal{P}\}$ is eventually a single target $\bar{q}$. Apply Lemma 5 to each coordinate, with $b_k = \alpha_k \mu_k(s)$ and zero errors, to obtain $q_k \rightarrow \bar{q}$. A policy occurring infinitely often is greedy for $\bar{q} = q_\pi$, and hence optimal by Lemma 2. ■

The common coefficient c is the reason for requiring equal action rates within each state. Different states may have different rates and may pause arbitrarily long, provided their cumulative learning diverges. Fixed-priority ties give the same deterministic result: an unchanged table retains the same priority maximizer, which is the only tie property used above.

3 A barrier on a state's learning clock

The same sampling factor appears in a gap's drift and variance. Keeping that factor permits arbitrarily long pauses: when a state is not sampled, both the restoring force and the noise pause. No divergence of the state's clock is needed for the following local estimate.

Lemma 9 (State-clock barrier) *Let (D_k) be a real adapted process with integrable increments, let $r \leq \tau$ be stopping times, and let $\sigma_k(s) \in [0, 1]$ be $\mathcal{F}'_k$ -measurable. The deterministic steps are in $(0, 1]$, square summable, and satisfy $\alpha_j \leq C_\alpha \alpha_k$ for all $j \geq k \geq k_\alpha$, with deterministic $C_\alpha \geq 1$ and k_α . Assume $r \geq k_\alpha$. Write*

$$m_k = \mathbb{E}[D_{k+1} - D_k \mid \mathcal{F}'_k], \quad Z_{k+1} = D_{k+1} - D_k - m_k.$$

Suppose deterministic $\delta, V > 0$ satisfy, on $\{r \leq k < \tau, D_k \geq 0\}$,

$$m_k \geq \alpha_k \sigma_k(s) (4\delta - D_k), \quad \mathbb{E}[Z_{k+1}^2 \mid \mathcal{F}'_k] \leq V \alpha_k^2 \sigma_k(s). \quad (16)$$

Then, on $\{r < \infty, D_r > 0\}$,

$$\mathbb{P}(\exists \text{ finite } n \in [r, \tau] : D_n \leq 0 \mid \mathcal{F}'_r) \leq C_0 \frac{C_\alpha \alpha_r}{\min\{D_r, \delta\}} + \frac{C_1}{\delta^2} \sum_{k=r}^{\infty} \alpha_k^2, \quad (17)$$

where $C_0 = \max\{1, 20V(1 + (3\delta)^{-1})\}$ and $C_1 = 16V$. The crossing event includes the update ending at τ .

Proof. *Before reaching δ .* Put $a_r = C_\alpha \alpha_r$, so every future step is at most a_r . Suppose $0 < D_r < \delta$. If $D_r < a_r$, the first term in (17) is already at least one. Otherwise define

$$\nu = \tau \wedge \inf\{n \geq r : D_n \leq 0 \text{ or } D_n \geq \delta\}, \quad M_n = \sum_{k=r}^{n-1} \mathbf{1}_{\{k < \nu\}} Z_{k+1},$$

and the state clock

$$A_s(r, n) = \sum_{k=r}^{n-1} \alpha_k \sigma_k(s). \quad (18)$$

A first crossing at $n \leq \tau$ before reaching δ forces

$$M_n \leq -D_r - 3\delta A_s(r, n). \quad (19)$$

For $b > 0$ known at r , set $T(b) = \inf\{n \geq r : A_s(r, n) \geq b\}$, possibly infinite. The bound $\alpha_k \leq a_r$ controls the clock overshoot and future variance, at every finite horizon $n \geq r$, by

$$A_s(r, T(b) \wedge n) \leq b + a_r, \quad \sum_{k=r}^{T(b) \wedge n-1} \alpha_k^2 \sigma_k(s) \leq a_r(b + a_r).$$

The martingale maximal inequality therefore gives (with the harmless factor 4)

$$\mathbb{P}\left(\sup_{r \leq n \leq T(b)} |M_n| \geq z \mid \mathcal{F}'_r\right) \leq \frac{4V a_r(b + a_r)}{z^2}, \quad z > 0. \quad (20)$$

The supremum is over finite indices. First stop at a finite horizon, then increase the horizon; an unattained supremum is handled by increasing thresholds to z from below. For $\mathcal{F}'_r$ -measurable b, z , localize on sets where b, z, z^{-1} are bounded. Random r is treated on $\{r = j\}$ and then combined over integers j . Thus no finite bound on $T(b)$ or the waiting time has been assumed.

Partition possible crossing times into $A_s(r, n) < D_r/(3\delta)$ and, for $j \geq 1$,

$$2^{j-1} D_r/(3\delta) \leq A_s(r, n) < 2^j D_r/(3\delta).$$

Apply (20) with $(b, z) = (D_r/(3\delta), D_r)$ for the first range and $(b, z) = (2^j D_r/(3\delta), 2^{j-1} D_r)$ for the others. Because $D_r \geq a_r$, the respective bounds are at most

$$4V(1 + (3\delta)^{-1}) \frac{a_r}{D_r}, \quad 16V(1 + (3\delta)^{-1}) \frac{a_r}{D_r} 2^{-j}.$$

Summing and using $a_r = C_\alpha \alpha_r$ gives the first term in (17).

After reaching δ . Let $\rho = \inf\{n \geq r : D_n \geq \delta\}$, using $\rho = r$ if $D_r \geq \delta$. On $\{\rho < \infty, \rho \leq \tau\}$, restart at ρ and stop at $\eta = \tau \wedge \inf\{n \geq \rho : D_n \leq 0\}$:

$$\widetilde{M}_n = \sum_{k=\rho}^{n-1} \mathbf{1}_{\{k < \eta\}} Z_{k+1}.$$

For $D_k \geq \delta$, (16) and $0 \leq \alpha_k \sigma_k(s) \leq 1$ imply $D_k + m_k \geq \delta$. If a first subsequent crossing is at n , let $\ell < n$ be the last index with $D_\ell \geq \delta$. All subsequent gaps before n are in $(0, \delta)$ and have nonnegative mean increments, so

$$D_n \geq \delta + \widetilde{M}_n - \widetilde{M}_\ell.$$

A crossing forces $\sup_{n \geq \rho} |\widetilde{M}_n| \geq \delta/2$. Doob's inequality at ρ bounds this probability by $16V\delta^{-2} \sum_{k \geq \rho} \alpha_k^2$. Multiply by the $\mathcal{F}'_\rho$ -measurable event $\{\rho < \infty, \rho \leq \tau\}$, average back to $\mathcal{F}'_r$, and use $\rho \geq r$. This proves the second term. The last-visit index ℓ is used only in a pathwise inequality; we never condition at it. Both parts retain the increment $k = \tau - 1$ and therefore protect the gap through the endpoint itself. $\blacksquare$